

HARSH PAWAR

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PROFESSIONAL PROFILE

Automotive engineer with approximately six years of industry experience, currently completing an MSc in Automotive Engineering with Electric Vehicles at Oxford Brookes University with a specialisation in noise, vibration and harshness, structural dynamics and vibration attenuation. Experienced in ANSYS Mechanical, MATLAB and Simulink for finite element modelling, modal analysis, harmonic response analysis and frequency response functions. Practical background in vehicle systems, structural damage assessment, repair procedures and technical investigation, with a track record of translating detailed technical analysis into clear engineering decisions communicated to technicians, engineers and senior stakeholders.

TECHNICAL SKILLS

NVH and Structural Dynamics

Modal analysis, harmonic response analysis, frequency response functions, structural resonance, mode shapes, vibration attenuation, tuned vibration absorbers, damping, order tracking, ride comfort, signal processing, vibroacoustic analysis, structural dynamics.

CAE and Simulation

ANSYS Mechanical, finite element modelling, Hyper Mesh, LS DYNA, MSC Adams, GT SUITE, GT Drive, MATLAB, Simulink.

Vehicle Engineering

Vehicle dynamics, automotive structures, powertrain systems, electric and hybrid vehicles, structural damage assessment, repair methodology, mechanical and electrical fault analysis, root cause investigation, vehicle system integration.

CAD and Engineering Tools

SolidWorks, Siemens NX basics, Microsoft Excel, technical documentation, engineering reporting.

EDUCATION

MSc Automotive Engineering with Electric Vehicles

Oxford Brookes University, Sept 2025 - September 2026

Relevant modules: Noise, Vibration and Harshness; Advanced Vehicle Dynamics; Advanced Powertrain Engineering; Electric Vehicles; Composite Design and Impact Modelling.

Bachelor of Engineering in Automobile Engineering; 2016 - 2019, First Class.

Diploma in Automobile Engineering; 2013 - 2016, First Class.

PROFESSIONAL EXPERIENCE

Motor Loss Assessor | Bajaj Allianz General Insurance | July 2019 to October 2024

- Contributed to increasing the quarterly claim settlement ratio at a high-end vehicle location from 81 percent to 97 percent through faster technical assessment, stronger workshop coordination and structured claim follow up.
- Assessed approximately 120 to 150 motor claims per month, evaluating vehicle damage, structural repair requirements, OEM repair procedures, parts, labour, technical evidence and repair estimates.
- Investigated complex claims, repair discrepancies and root causes, supporting approximately 30 to 35 associated workshops and garages and coordinating with technicians, customers and internal stakeholders.
- Trained and supported new motor loss assessors while balancing repair quality, safety, cost control, policy compliance and customer outcomes.

Training and Compliance Officer | Toyota and MG Dealerships | October 2024 to August 2025

- Increased the proportion of trained technicians from 35 percent to 67 percent within four months across participating Toyota and MG dealerships through structured technical, process and compliance training.
- Improved vehicle inspection quality and safety compliance performance by 17 percent through OEM procedure review, workshop audits, corrective action planning and structured follow up.
- Delivered technical training to technicians and service advisers across several dealership branches, reviewing diagnostics, vehicle inspection processes and safety requirements, with awareness of electric vehicle service procedures.

Engineering Intern | Ford Motors | June 2015 to February 2016

- Diagnosed mechanical and electrical vehicle faults using OEM diagnostic tools and supported repair verification through quality control inspections and road testing, helping reduce the risk of incomplete or incorrect repairs before final vehicle release.

Powertrain and Electronics, Oxford Brookes Racing (Formula Student)

Formula Student Artificial Intelligence team, Oxford Brookes University, 2025 to present

- Worked within the Powertrain team on CAD corrections and design failure mode and effects analysis (DFMEA).
- Worked within the Electronics and Vehicle Controls team, helping build the shutdown circuit and PCBs for vehicle control systems.
- Competed at Formula Student UK 2026 across the Formula Student AI and Artificial Perception Challenge classes; the team won the Driverless Dynamic Task class and finished second overall in the Artificial Perception Challenge car category.

ENGINEERING PROJECTS

YASA Motors Vibroacoustic Modelling: Application of Metamaterial Concepts to Reduce Vibration of a Motor Cover

Ongoing MSc dissertation, Oxford Brookes University

- Built a staged beam to plate to motor cover simulation workflow in ANSYS Mechanical, running modal and harmonic response analyses to generate frequency response functions.
- Identified structural resonances, anti-resonances and attenuation regions, targeting the 80 Hz to 130 Hz range at beam stage before extending to a YASA axial flux motor cover across approximately 100 Hz to 1000 Hz.
- Investigating locally resonant metamaterial concepts and tuned resonators, evaluating arm length, geometry, tip mass, position and tuning frequency for frequency selective vibration reduction, with physical validation planned.

Noise, Vibration and Harshness Engineering Portfolio

- Analysed ride comfort, modal testing, vibration absorber design and rotating machinery order tracking using MATLAB, Simulink and ANSYS based finite element methods.

Composite Battery Enclosure: Laminate Design and Impact Optimisation

- Designed a 15-ply composite EV battery enclosure in Hyper Mesh and LS DYNA, reducing front impact intrusion from 51.4 mm to 5.0 mm and peak stress by 77.9 percent.

REFERENCES

- Available upon request.